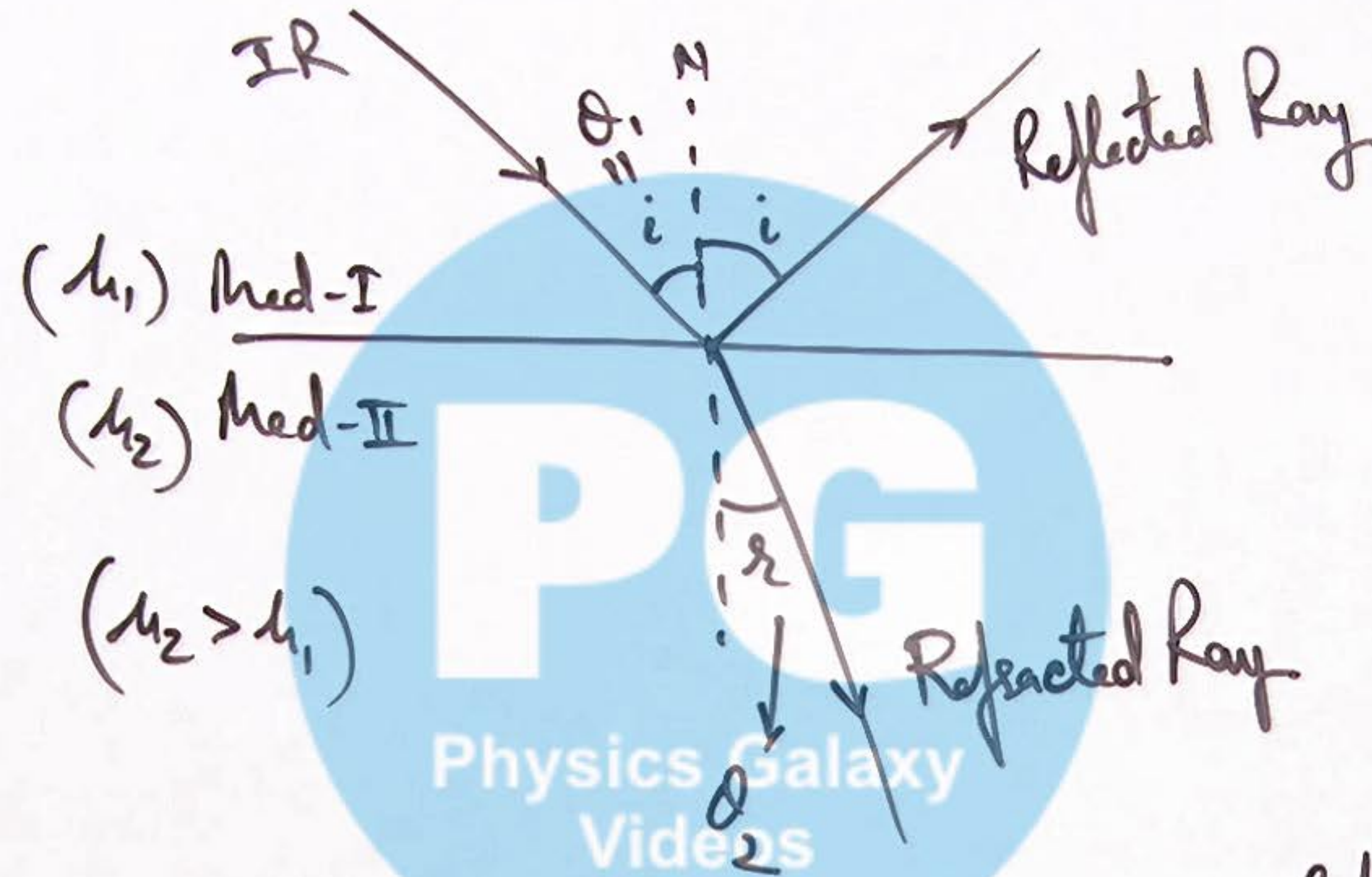


① Refraction of Light :

$$\left[\mu = \frac{c}{v} \right]$$



Laws of Refraction

① IR, RR & N lie in same plane.

② $\mu_1 \sin i_1 = \mu_2 \sin i_2$

$\mu_2 = \frac{\mu_2}{\mu_1} = \frac{\sin i_1}{\sin i_2}$
 Relative RI of med 2 w.r. to med 1

② Multiple Refractions Through Parallel Sided Slabs:

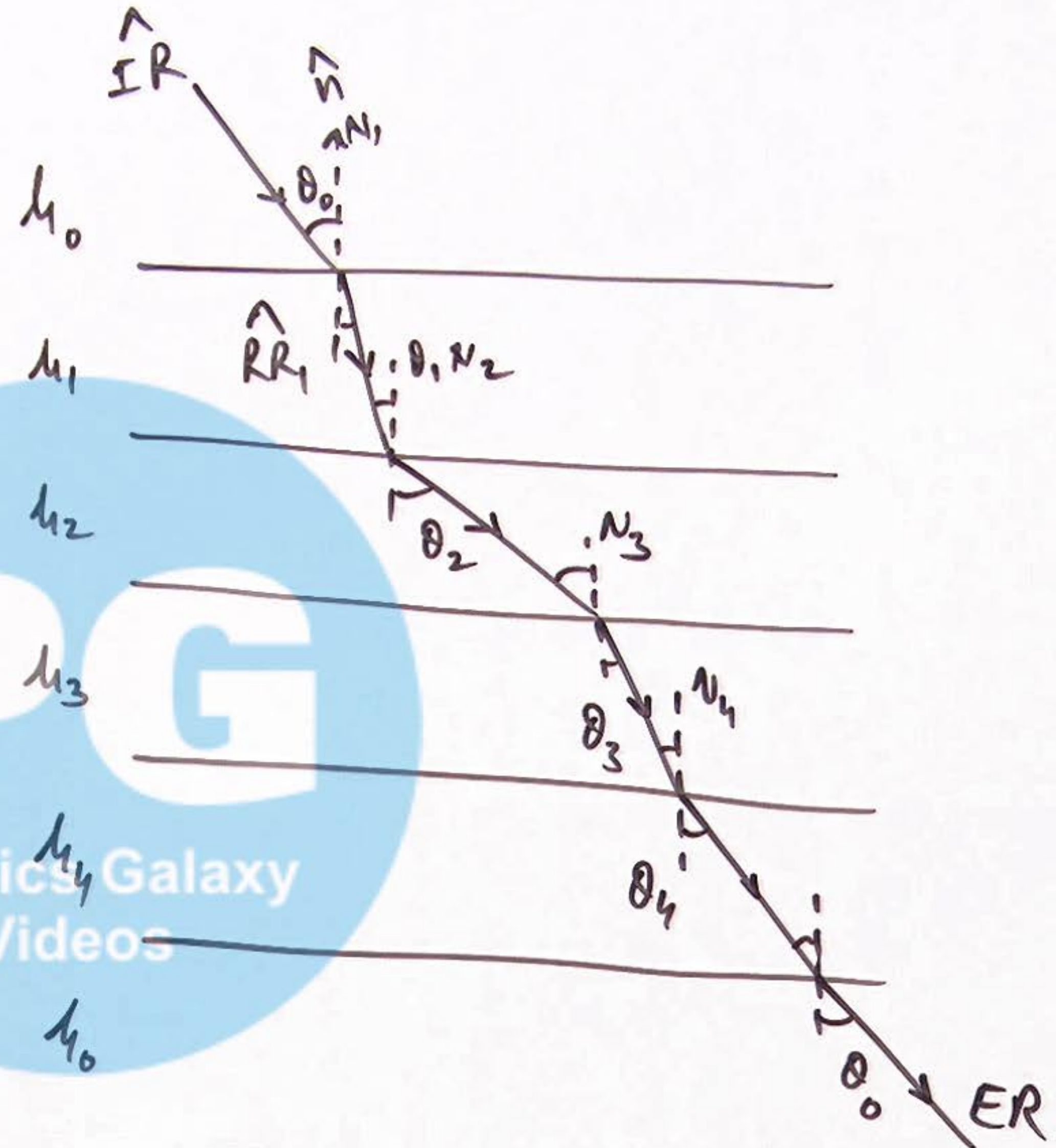
Snell's law

$$\mu \sin \theta = \text{Constant}$$

$$\mu_0 \sin \theta_0 = \mu_1 \sin \theta_1 = \mu_2 \sin \theta_2 = \dots$$

→ vector form,

$$\mu_1 (\hat{IR} \times \hat{n}) = \mu_2 (\hat{RR} \times \hat{n})$$



③ Image formation due to

Refraction at a Plane Surface : $\rightarrow$

★ Refraction formula.

$$\frac{u_1}{u} = \frac{u_2}{v}$$

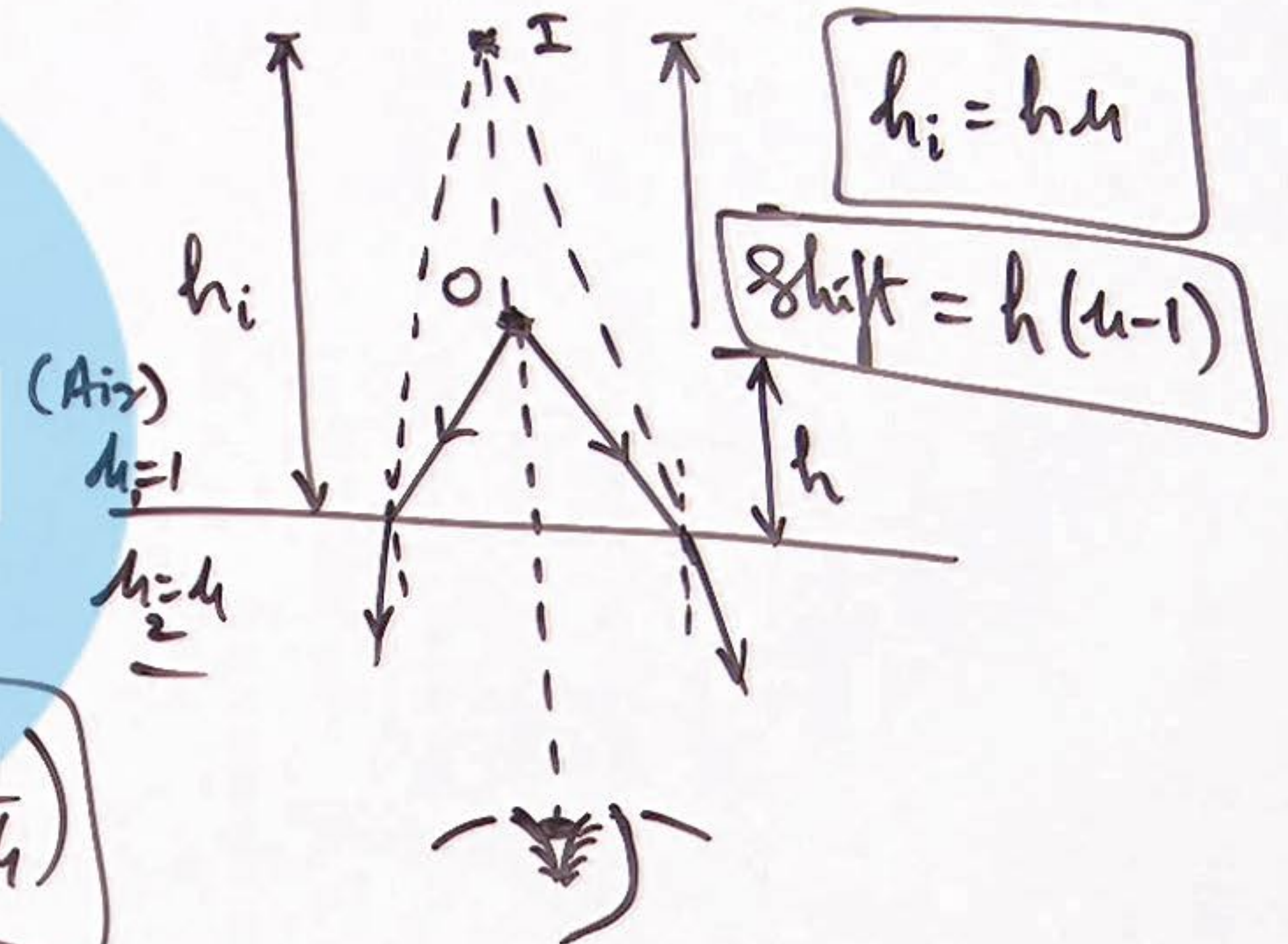
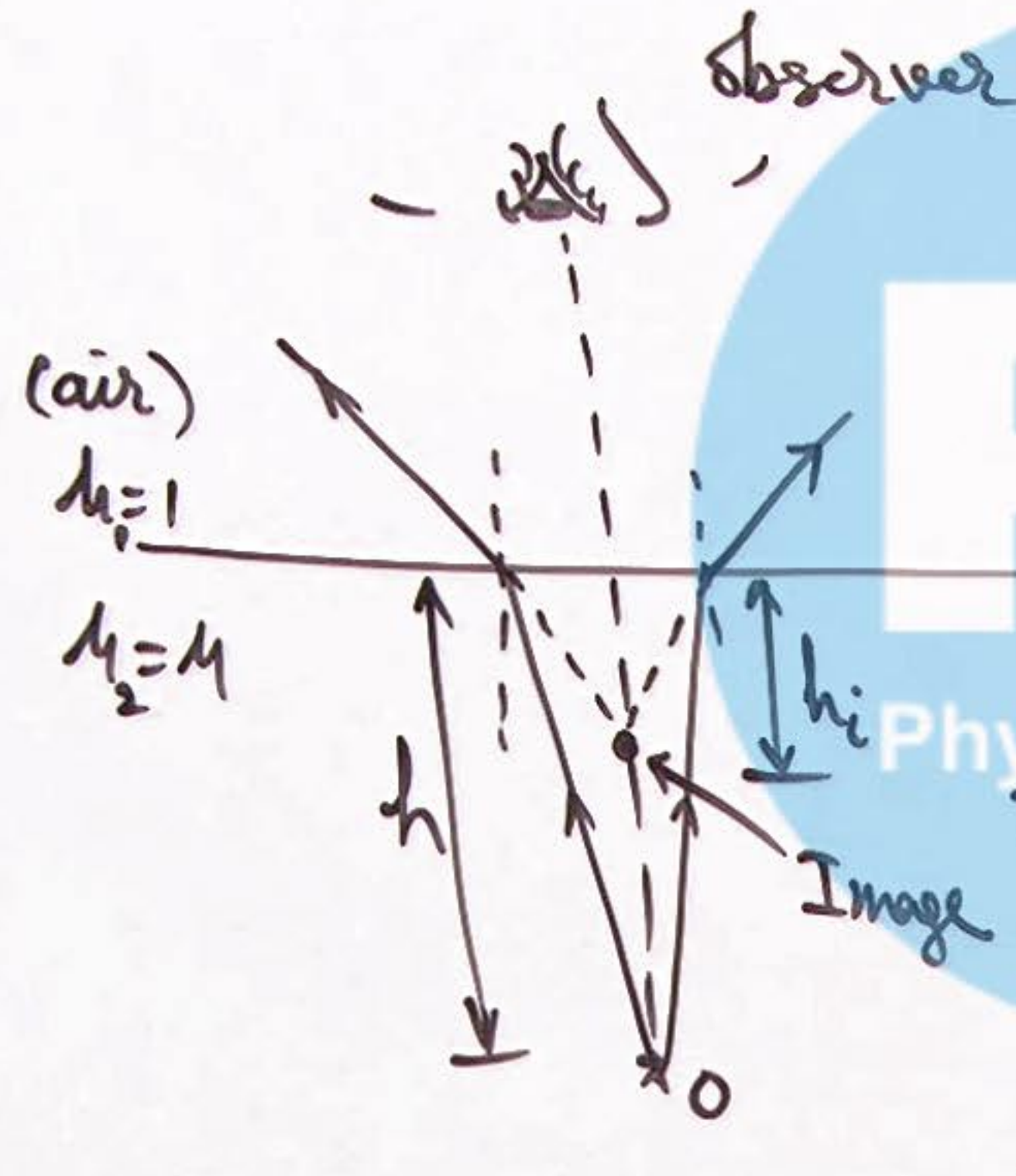
$u_1 \rightarrow$ RI of IR.
 $u_2 \rightarrow$ RI of RR

Case-1 : Object in denser medium

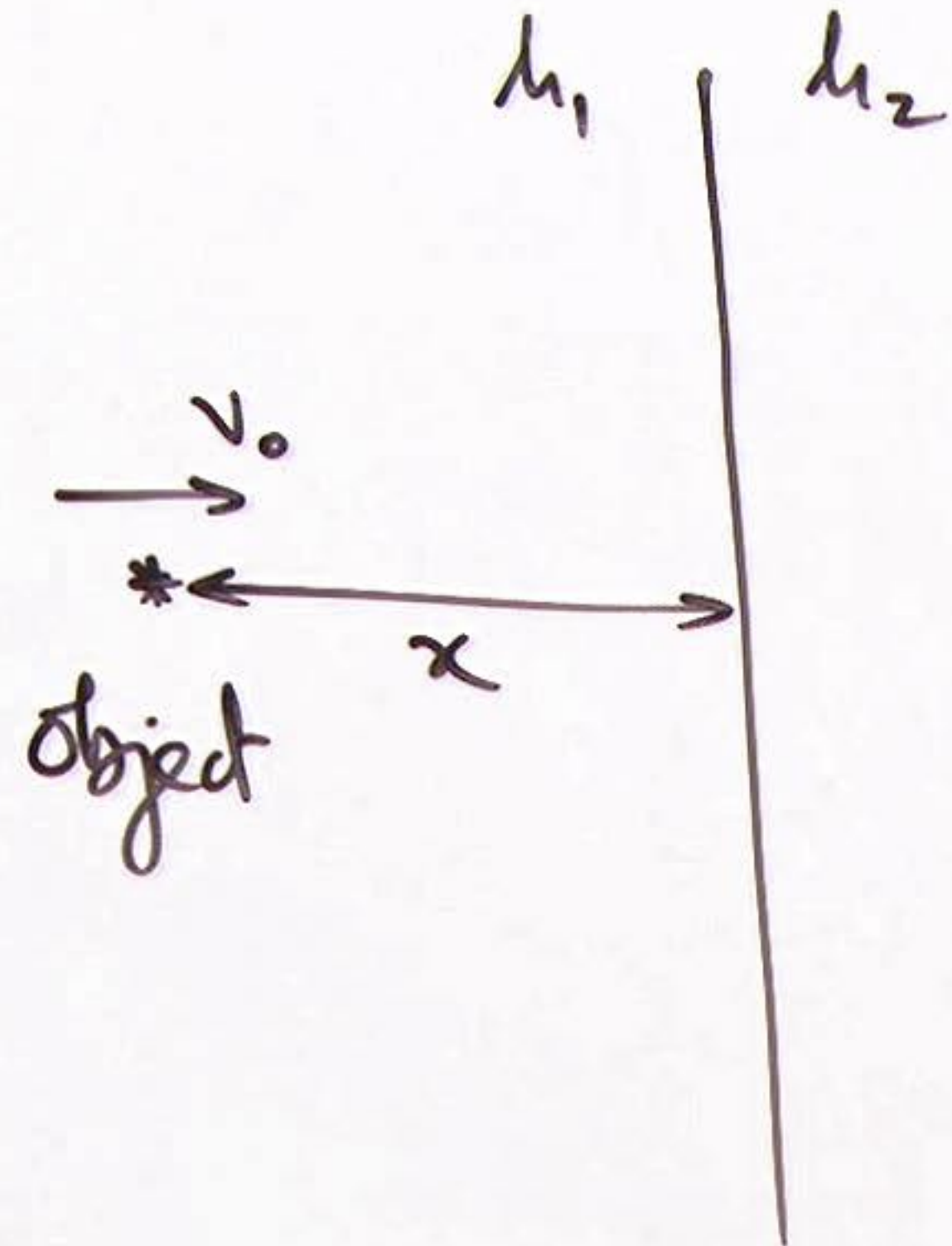
Case-2 : Object is in Rarer medium

$$h_i = \frac{h}{u}$$

Apparent depth.



④ Apparent Velocity due to Refraction :



$$\frac{n_1}{x} = \frac{n_2}{y}$$

$$\Rightarrow y = \frac{n_2}{n_1} \cdot x$$

differentiating w.r. to t

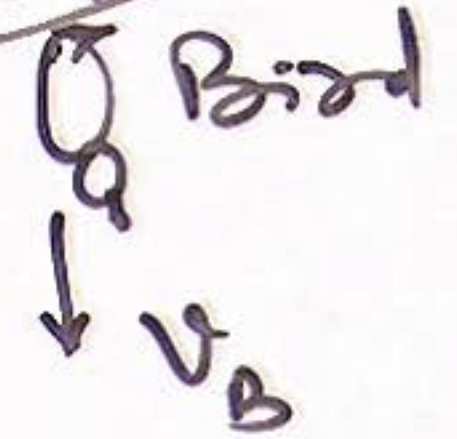
$$v_I = \frac{n_2}{n_1} \cdot v_o$$

$$v_{app} = v_I + v_i$$



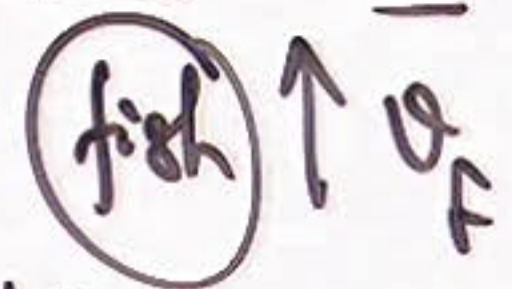
app vel of fish as seen by bird

$$v_{FB} = v_B + \frac{v_F}{n_w}$$



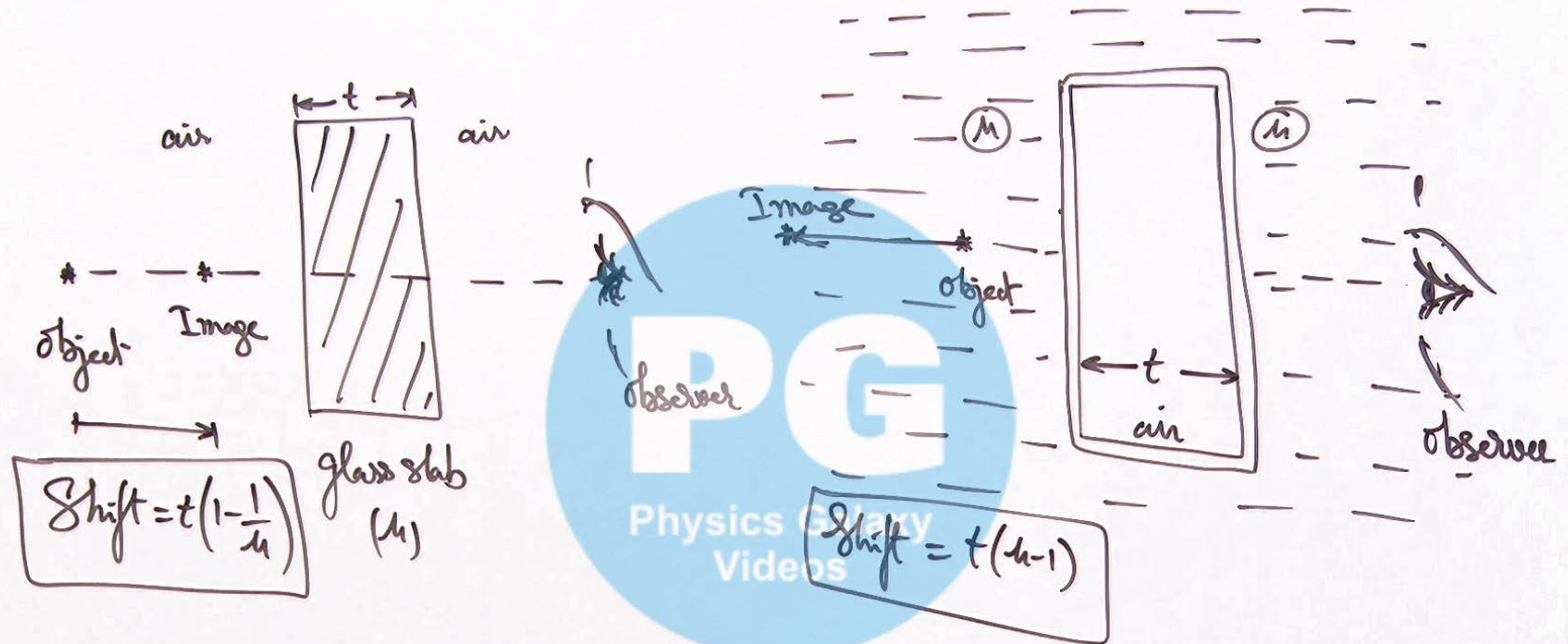
$$n_{air} = 1$$

$$n_w$$

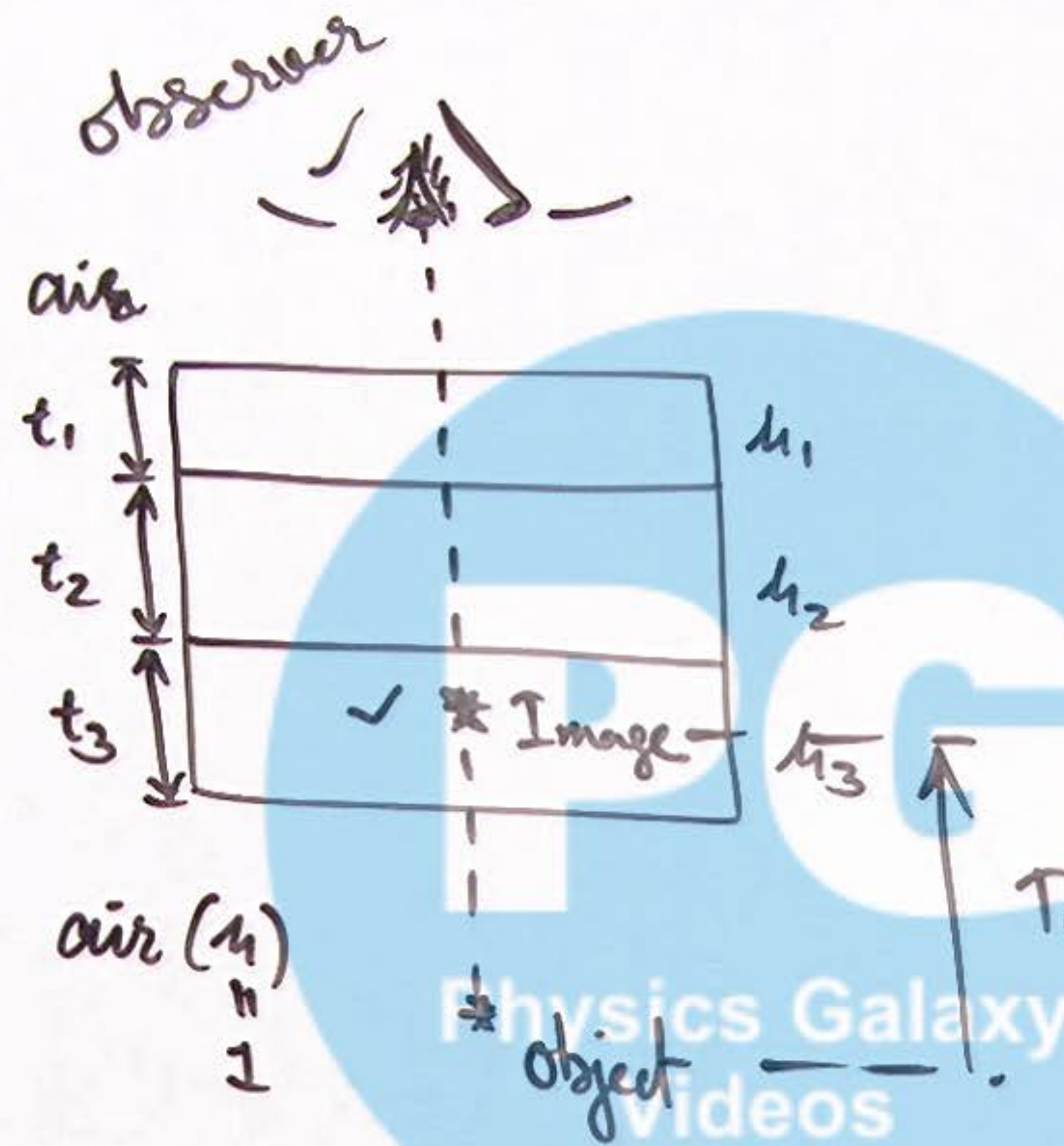


app vel of bird $\uparrow$ at v_B as seen by fish = $n_w v_B - v_F$

⑤ Image Shift by a Glass Slab:



⑥ Shift of object due to
Multiple Glass Slabs:



valid only for near normal incidence of light

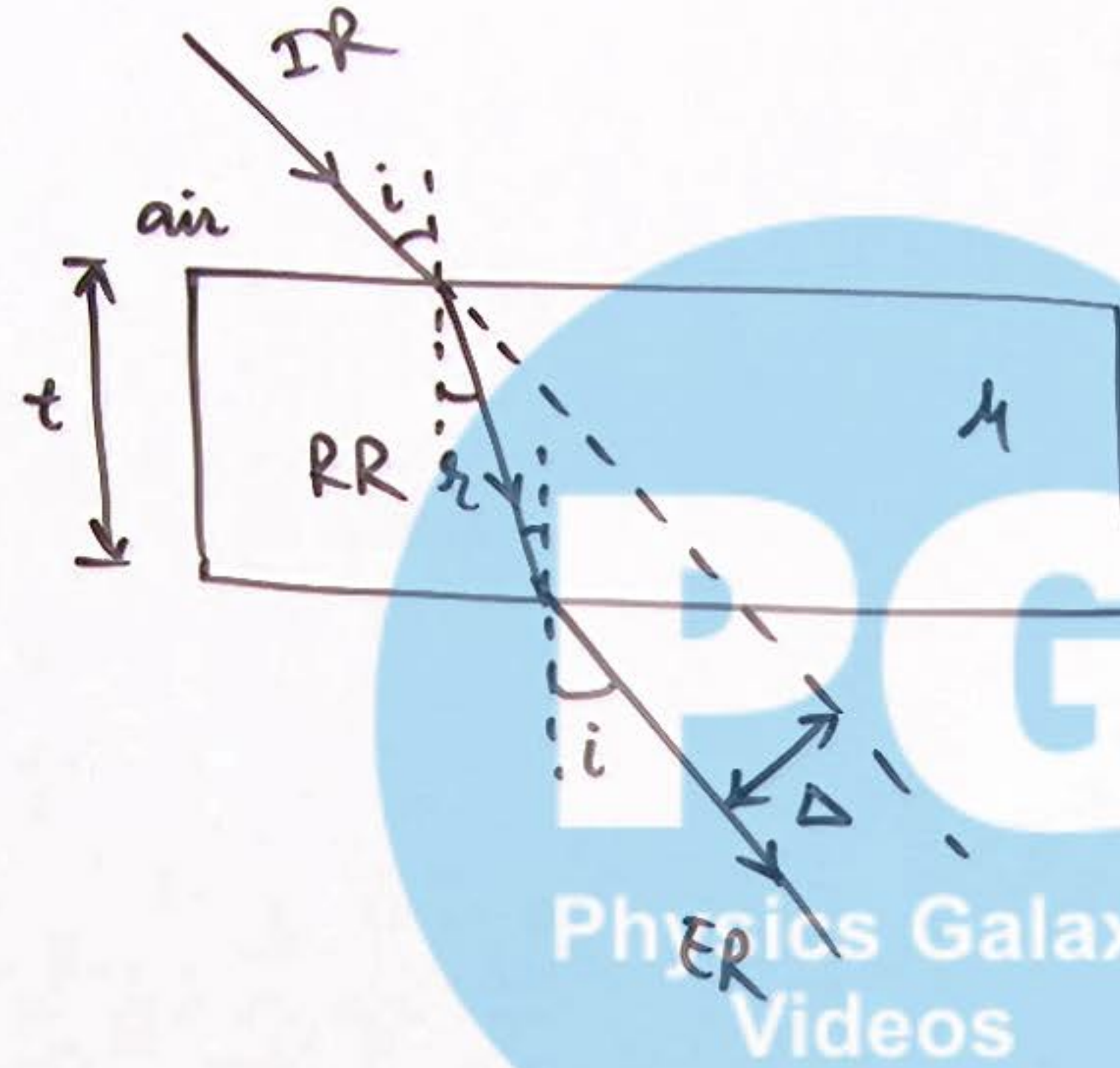
$$\text{Total Shift} = t_1 \left(1 - \frac{1}{n_1} \right) + t_2 \left(1 - \frac{1}{n_2} \right) + t_3 \left(1 - \frac{1}{n_3} \right)$$

⑦ Lateral Displacement by a Glass Slab:

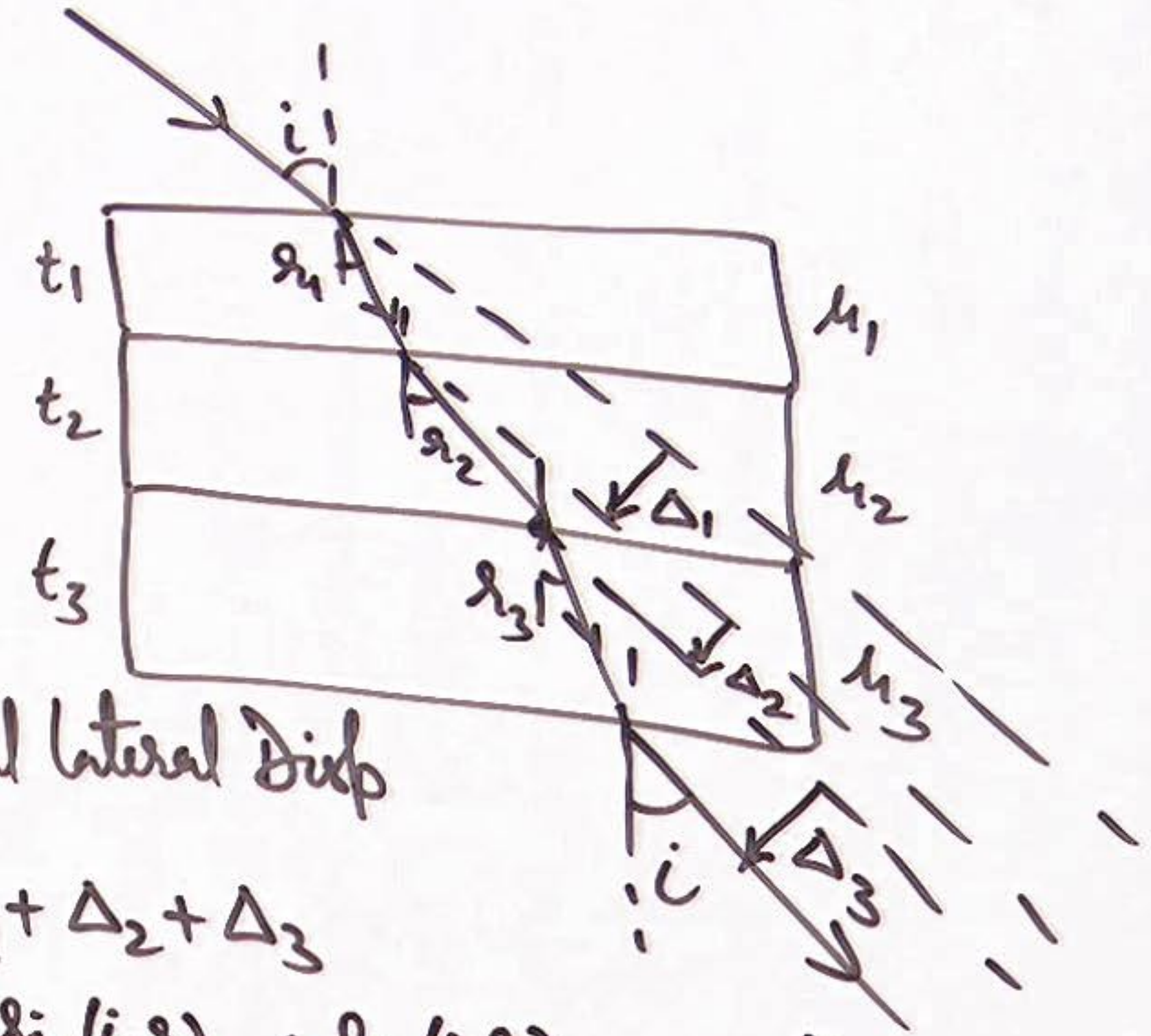
$$\sin i = \mu \sin r$$

lateral displacement

$$\Delta = \frac{t \sin(i-r)}{\cos r}$$



⑧ L.D. by Multiple Refractions by several slabs:

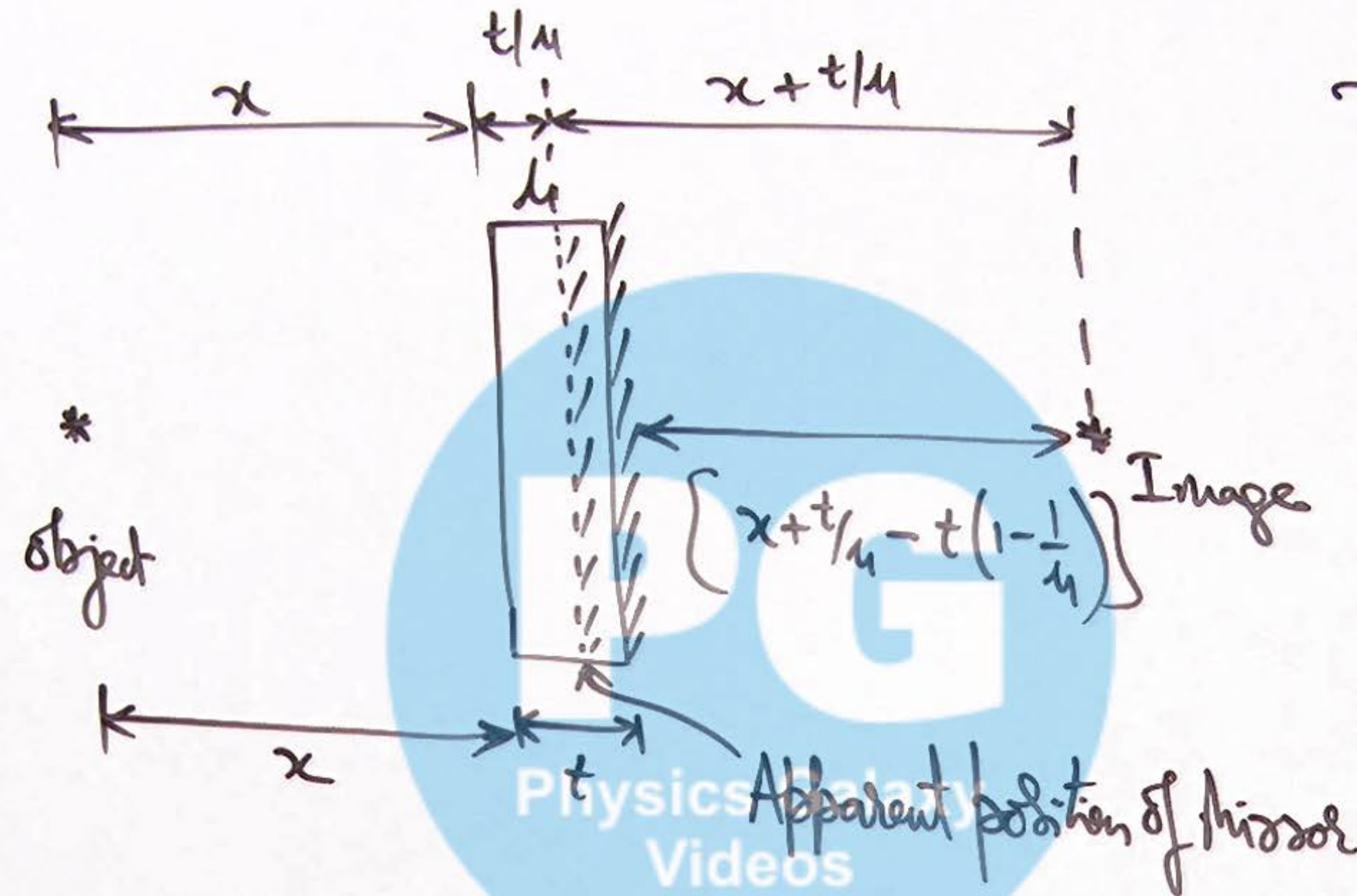


Total lateral Disp

$$\Delta_T = \Delta_1 + \Delta_2 + \Delta_3$$

$$\Delta_T = \frac{t_1 \sin(i-r_1)}{\cos r_1} + \frac{t_2 \sin(i-r_2)}{\cos r_2} + \dots$$

⑨ Image Formation by a Thick Mirror:



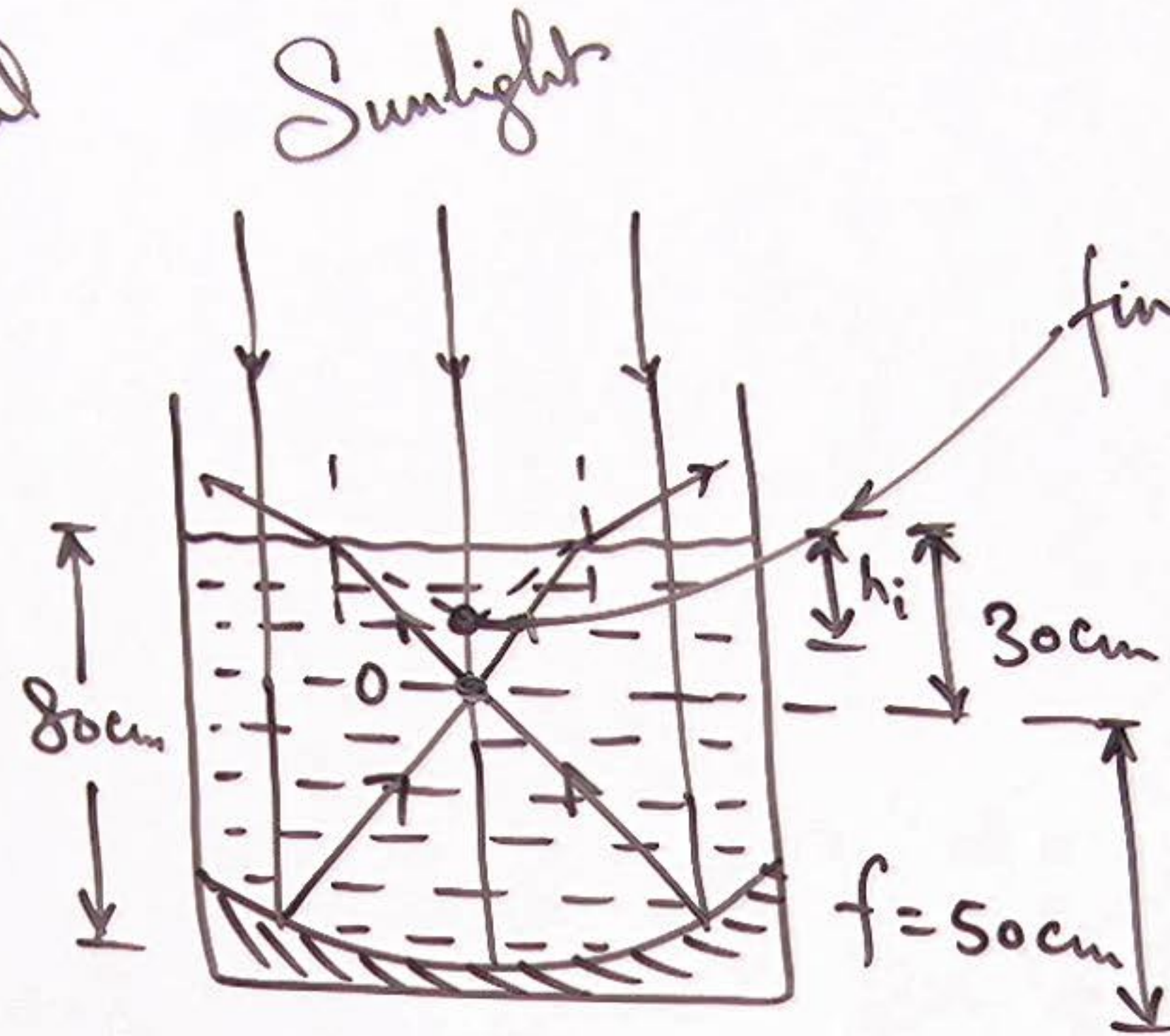
Distance between
O & I
 $= 2\left(x + \frac{t}{n}\right)$

10 Effect of medium on Reflection:

↓
"No"

IL: Find image produced by this optical system for sunlight.

$$\mu_w = 4/3$$



final image produced by sunlight

$$h_i = \frac{30}{\mu_w} = \frac{30}{4} \times 3$$

$$\Rightarrow h_i = \frac{90}{4} = 22.5 \text{ cm}$$

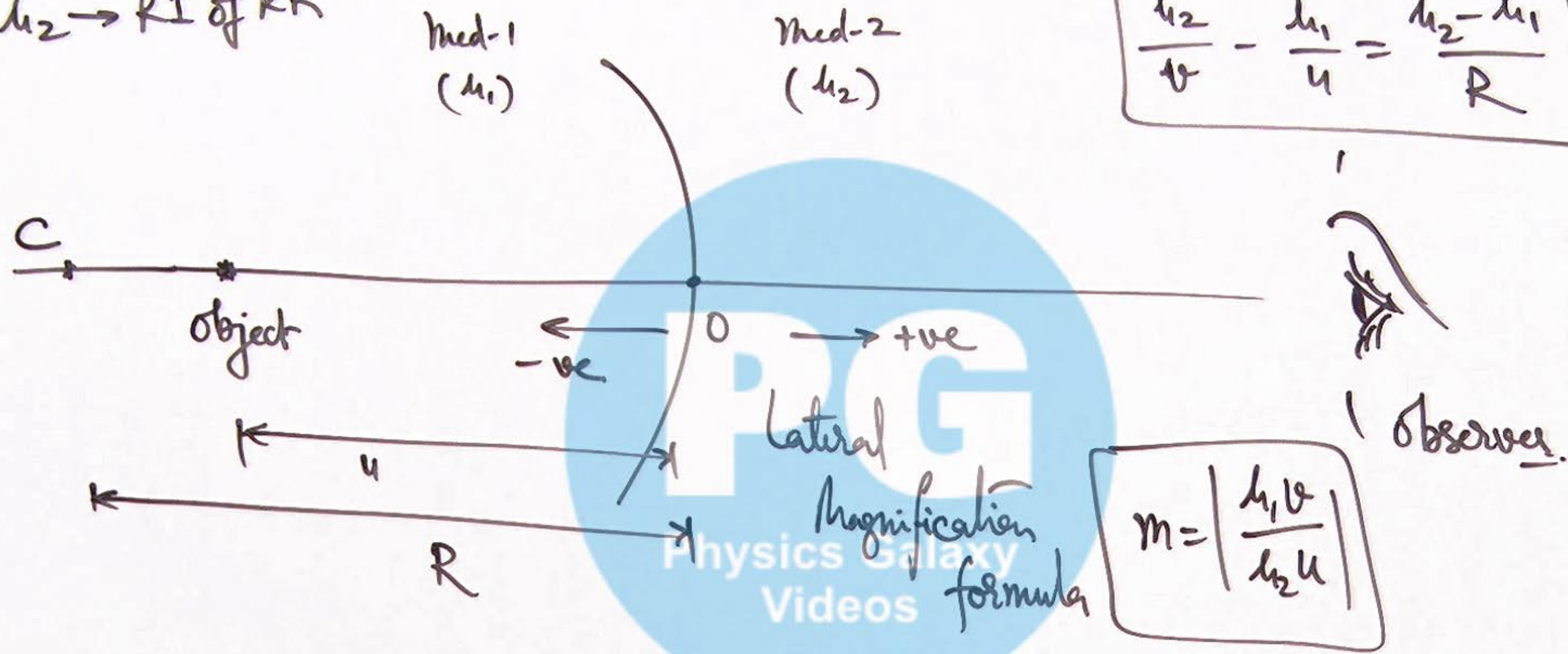
11) Refraction of light at Spherical Surfaces:

$\mu_1 \rightarrow$ RI of IR

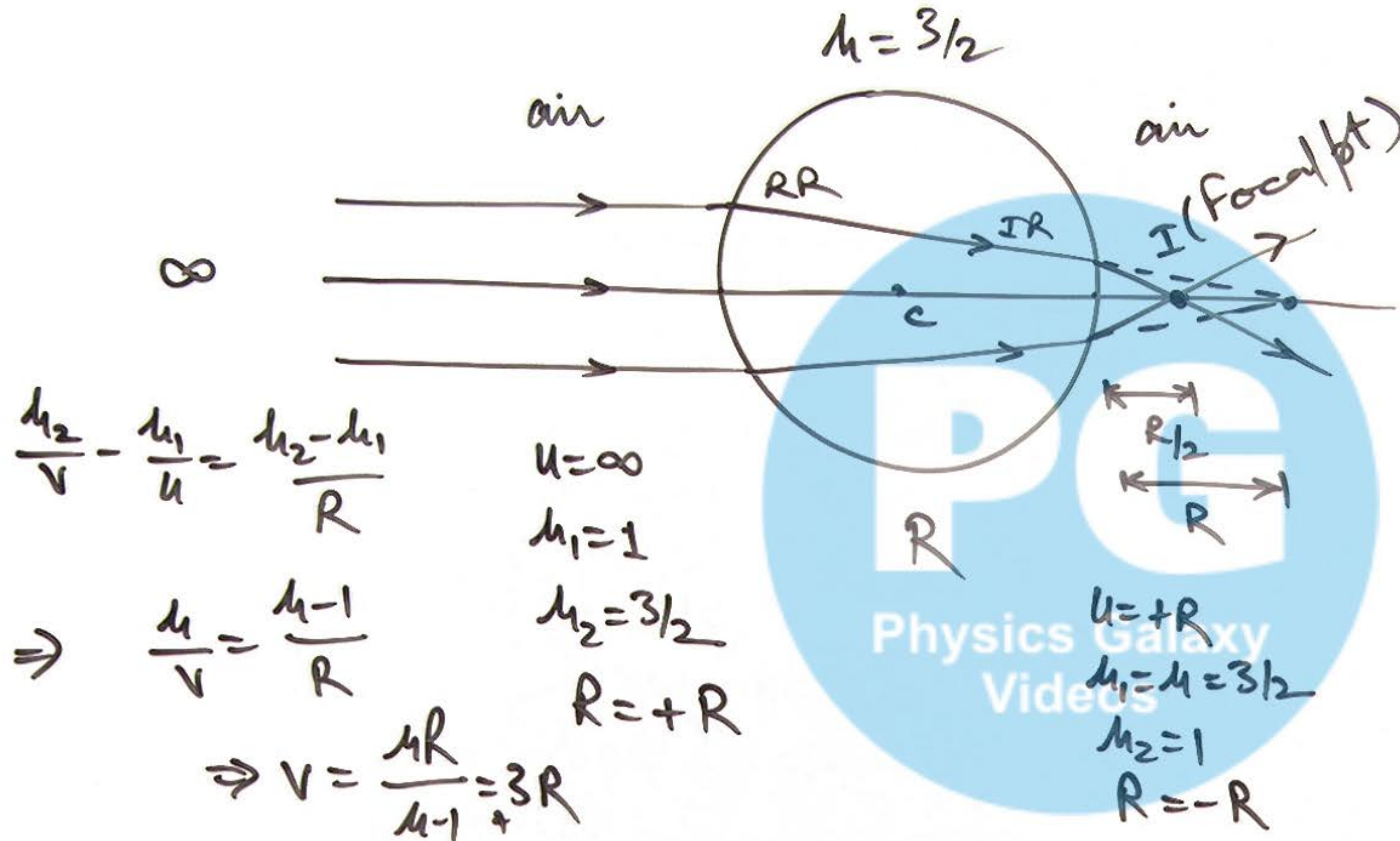
$\mu_2 \rightarrow$ RI of RR

Refraction Formula at Spherical Surface

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$



12 Focal length of a Glass Sphere:



$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

$$\frac{1}{v} - \frac{3}{2R} = \frac{1 - 3/2}{-R}$$

$$\Rightarrow \frac{1}{v} = \frac{3}{2R} + \frac{1}{2R} = \frac{2}{R} \Rightarrow v = +R/2$$

13) Refraction of a light Ray
through a glass sphere:



$$\delta = 2i - 2r$$

$$\checkmark \delta = 2i - 2 \sin^{-1} \left(\frac{\sin i}{\mu} \right)$$

$$\sin i = \mu \sin r$$

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⑭ Longitudinal magnification in Refraction :

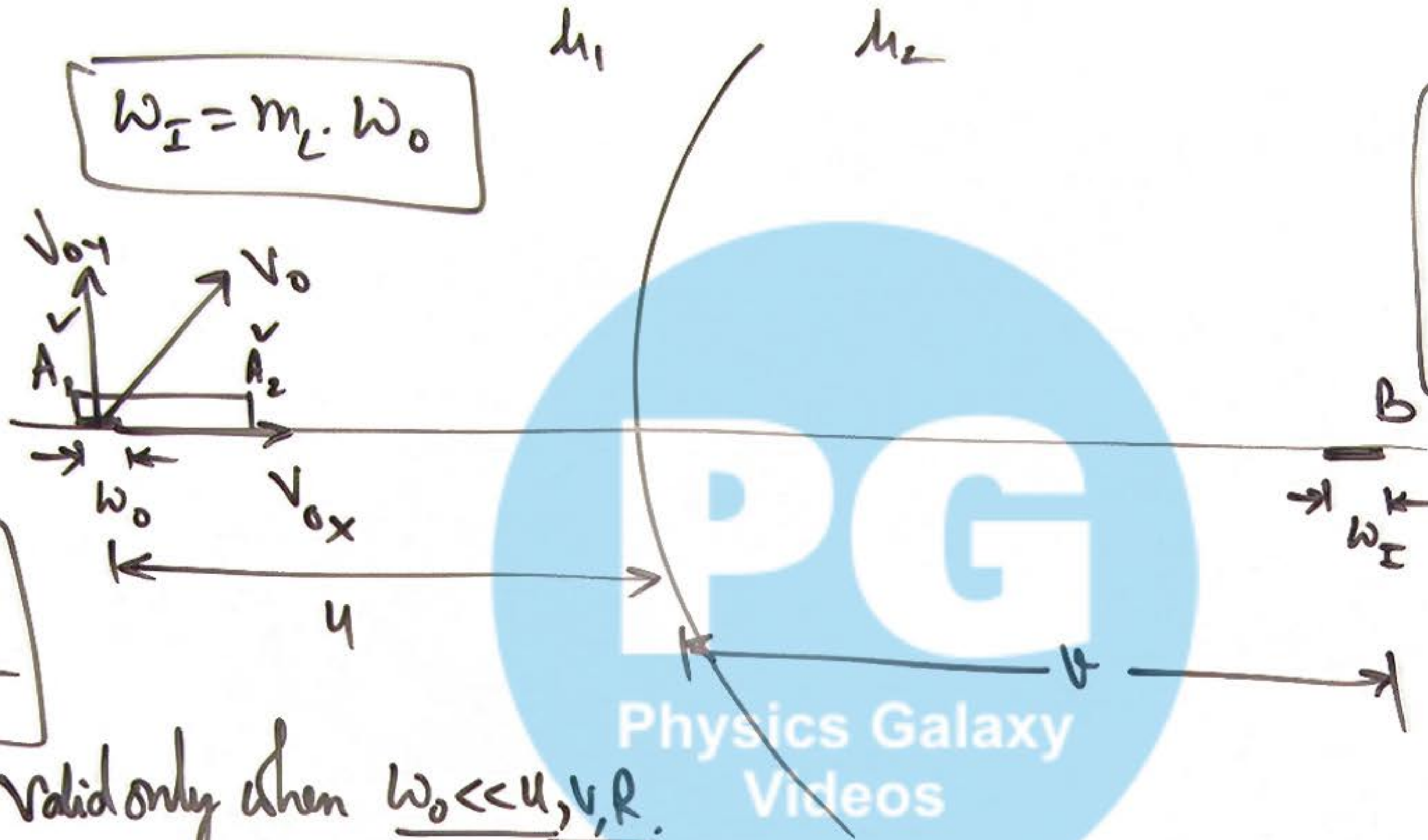
lateral mag

$$m = \frac{h_1 v}{h_2 u}$$

$$w_I = m_L \cdot w_o$$

$$m_L = \frac{h_1 v^2}{h_2 u^2}$$

valid only when $w_o \ll u, v, R$.



⑮ velocity of Image in Refraction :

$$V_{Iy} = m \cdot V_{oy}$$

$$\neq V_{Ix} = m_L \cdot V_{ox}$$

16 Total Internal Reflection (TIR):

It happens
when a light ray goes
from a denser to rarer medium.

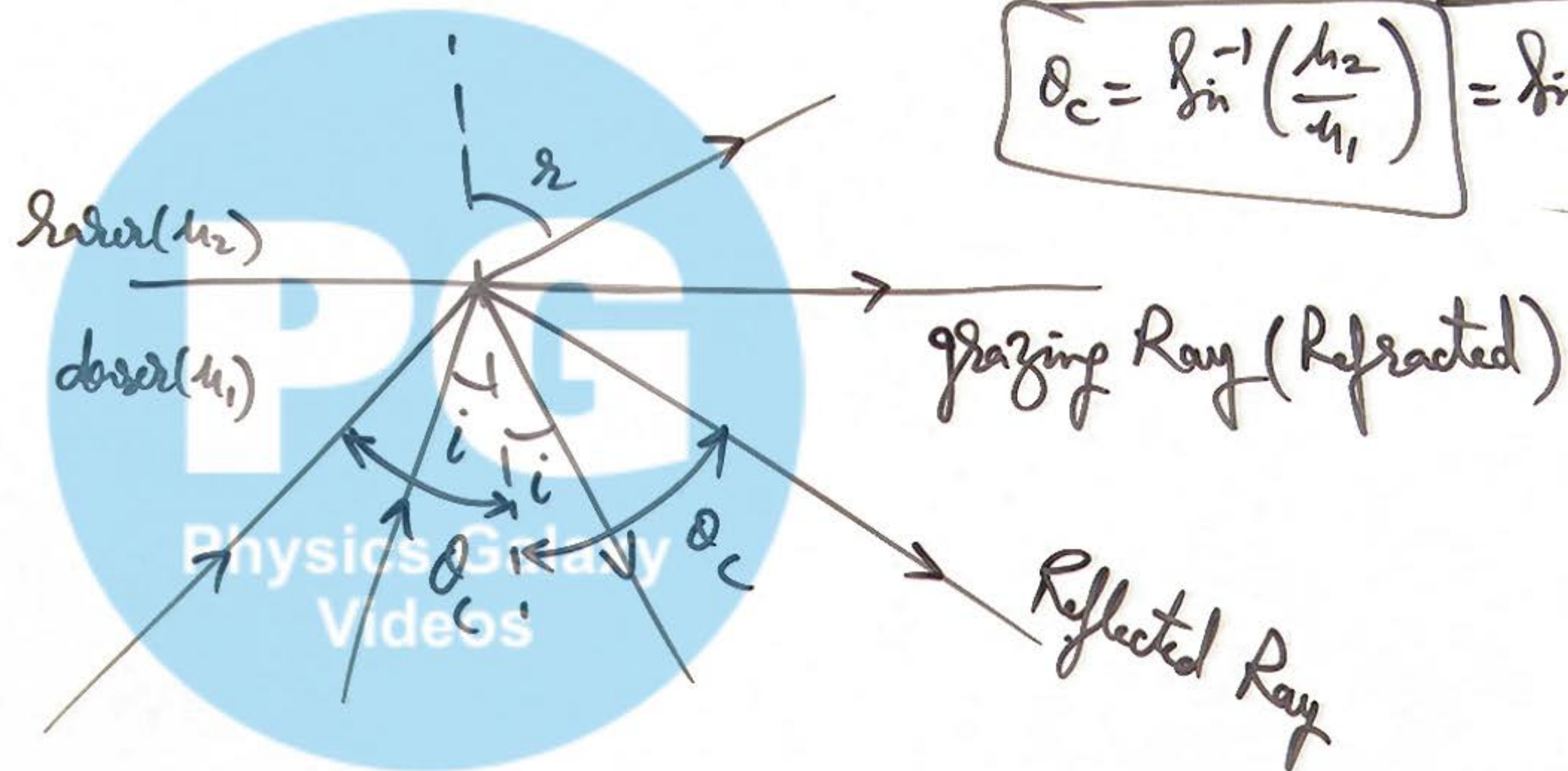
Using Snell's law

$$n_1 \sin \theta_c = n_2 \quad \text{for air} \downarrow$$

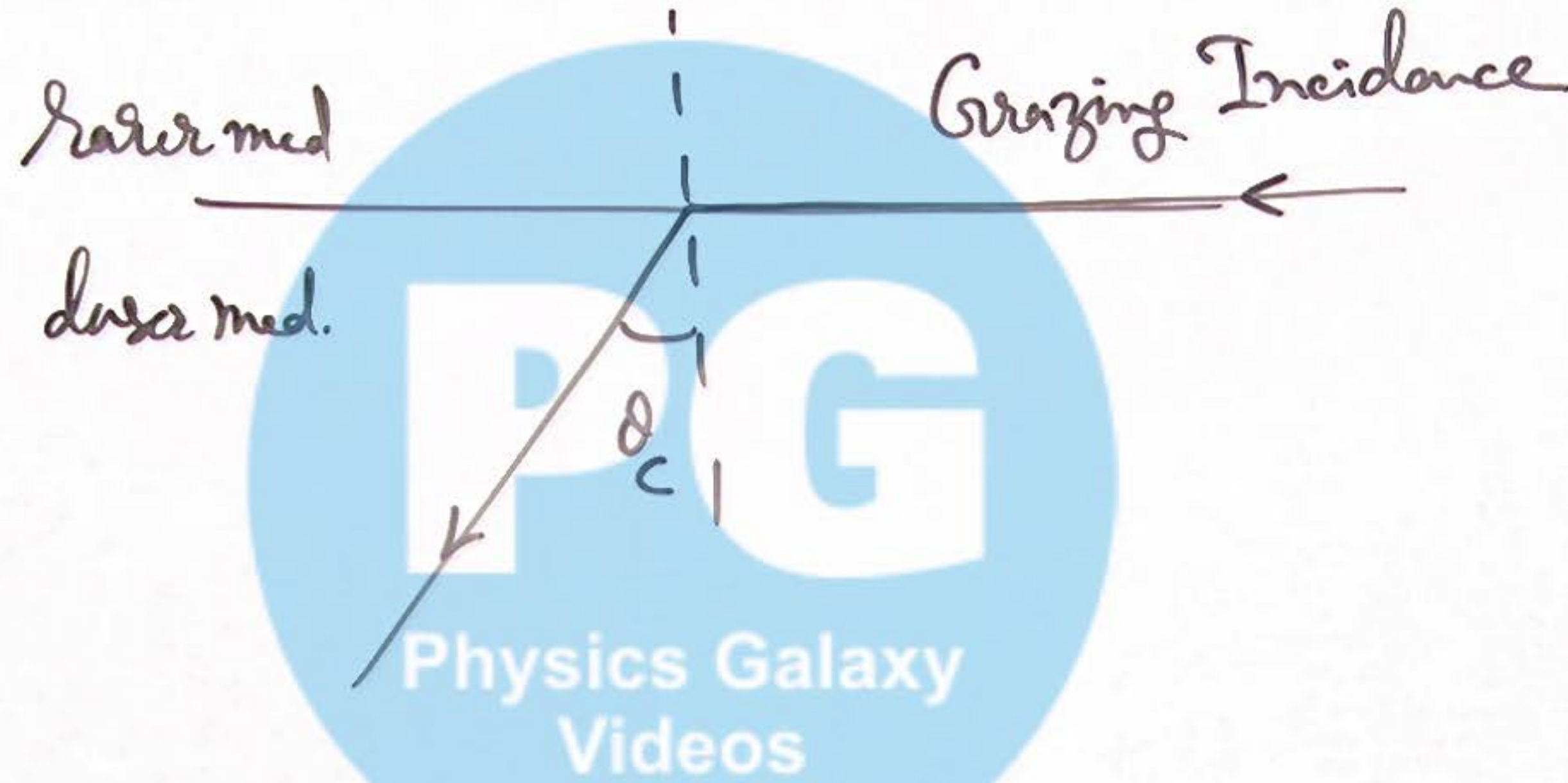
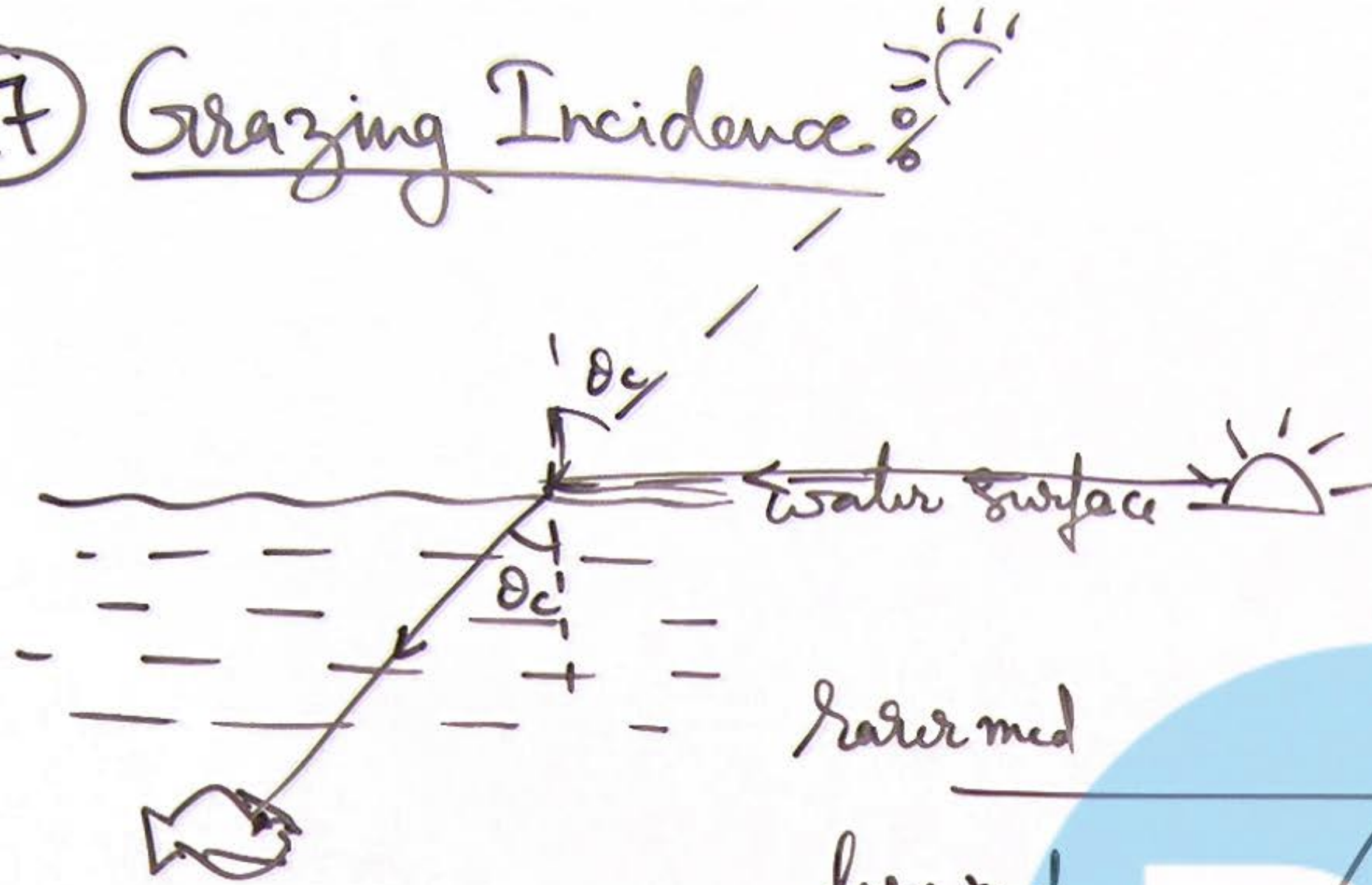
$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right) = \sin^{-1} \left(\frac{1}{n_1} \right)$$

$$n_2 < n_1$$

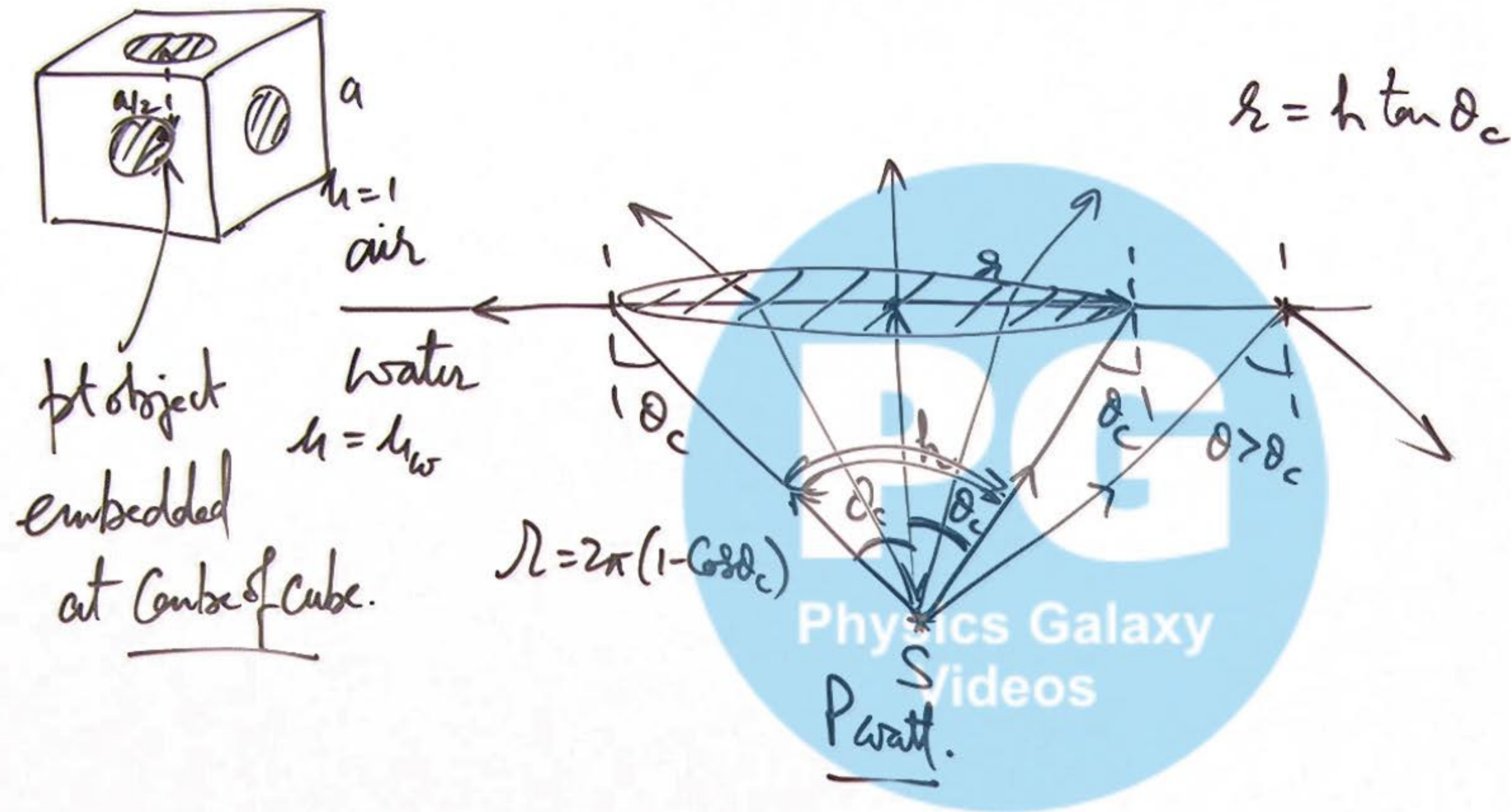
When i increases
even slightly beyond θ_c
then TIR takes place.



①7 Grazing Incidence



18 A point light source
inside a denser medium:



light energy
escaping water is

$$P_{es} = \frac{P}{4\pi} \times 2\pi(1 - \cos \theta_c)$$

$$P_{es} = \frac{P}{2} \left(1 - \frac{h}{\sqrt{h^2 + a^2}} \right)$$

19) Concept of Optical Fiber:

$$r_{\max} = \theta_c$$

min value of $90 - r$ is

$$90 - \theta_c \geq \theta_c$$

$$\Rightarrow \underline{\theta_c \leq 45^\circ}$$

$$\Rightarrow \theta_c = \sin^{-1}\left(\frac{1}{\mu}\right) \Rightarrow \boxed{\mu \geq \sqrt{2}}$$



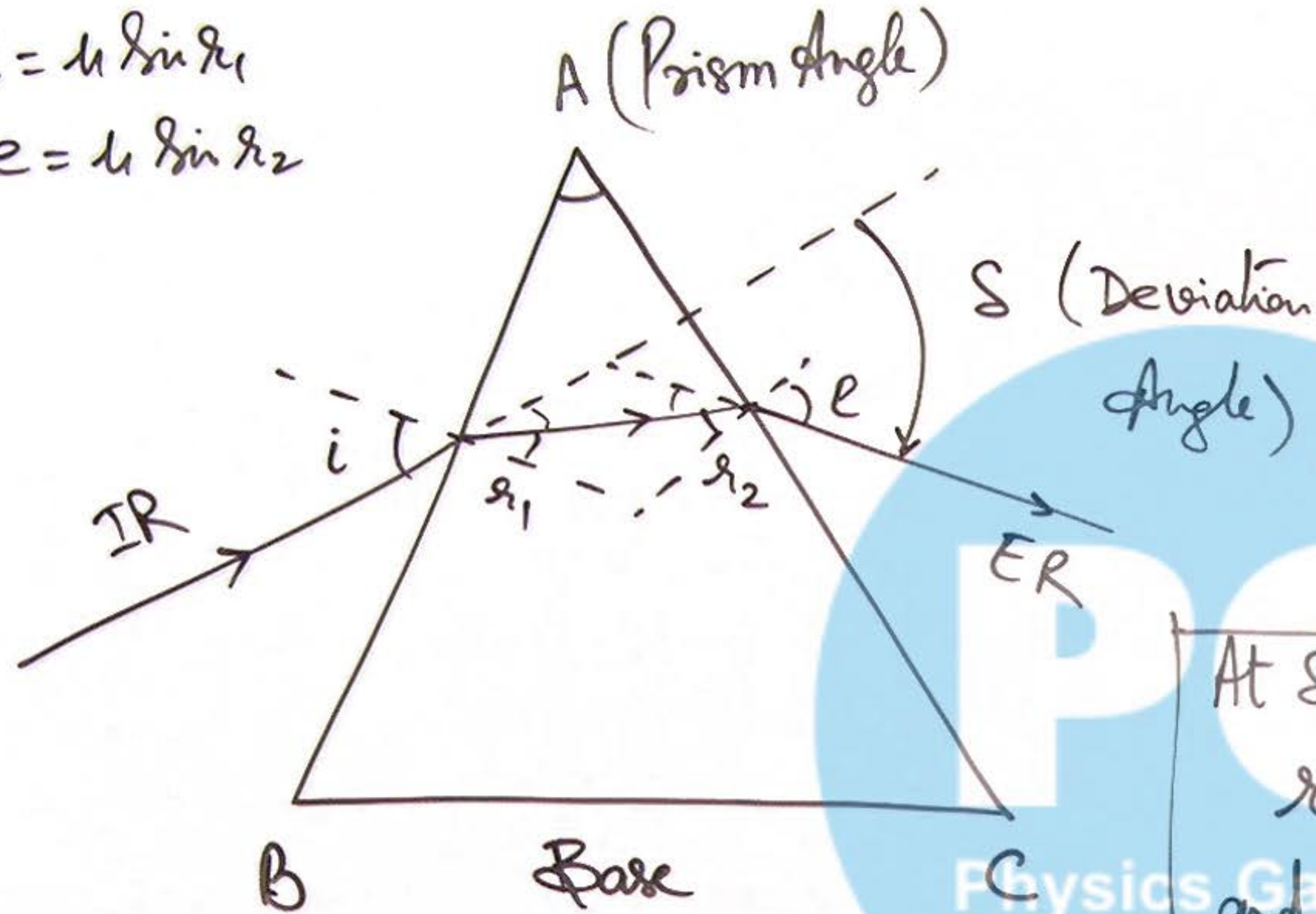
for optical fiber at every pt TIR we use $90 - r \geq \theta_c$

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20) Refraction through a Trihedral Prism:

$$\sin i = \mu \sin r_1$$

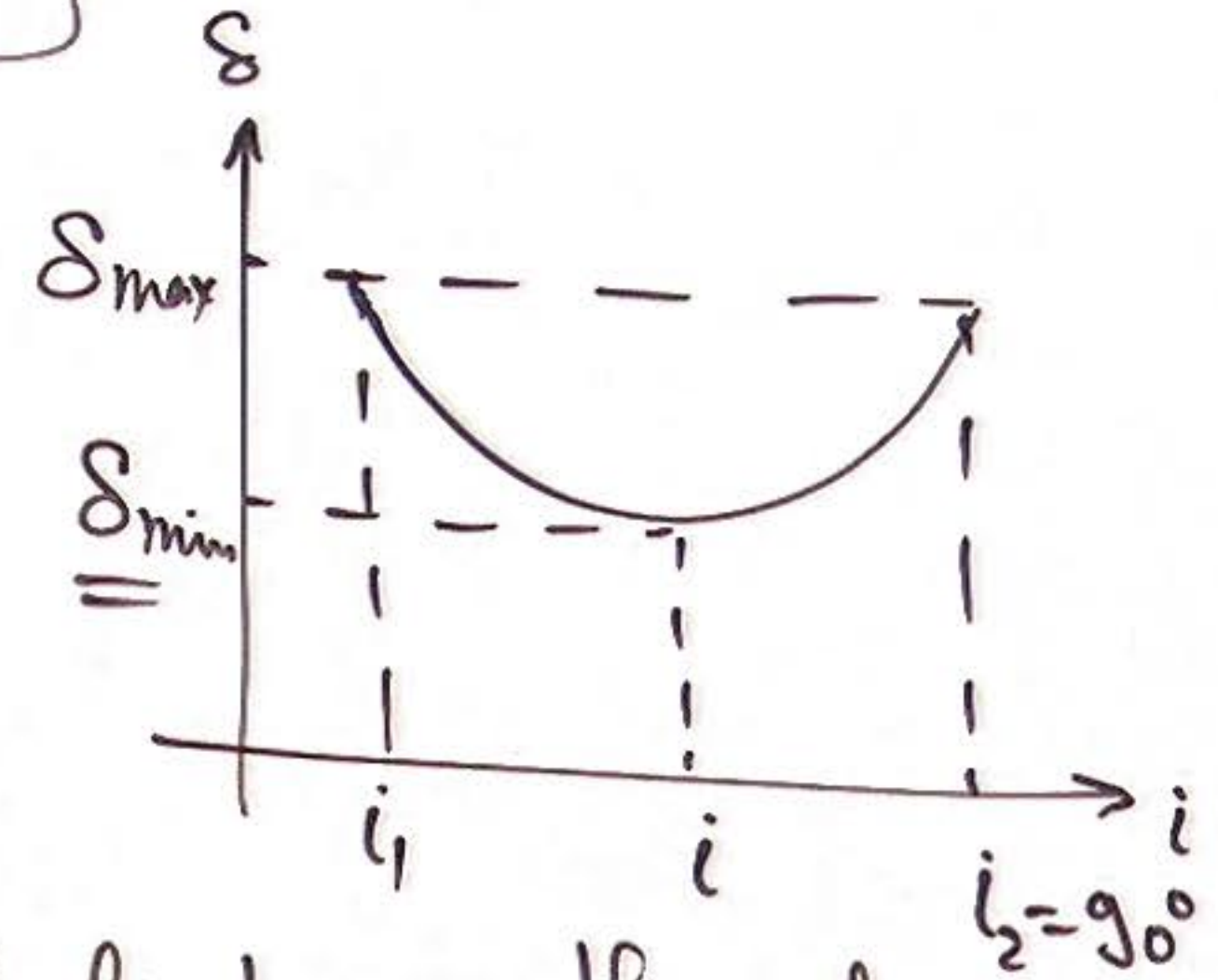
$$\sin e = \mu \sin r_2$$



$$r_1 + r_2 = A$$

$$\underline{S = i + e - A}$$

$$i + \sin^{-1}(\mu \sin(A - \sin^{-1}(\frac{\sin i}{\mu})) - A$$



At $S = S_{\min}$
 $r_1 = r_2 = A/2$
 and $i = e$

$S_{\min}$ is given by -

$$\sin \left[\frac{A + S_{\min}}{2} \right] = \mu \sin \left(\frac{A}{2} \right)$$

light passes through Prism Symmetrically

22 Condition for a light to pass through a Prism :

Cond 1 $\rightarrow A < 2\theta_c$

Cond 2 $\rightarrow i$ should be such that

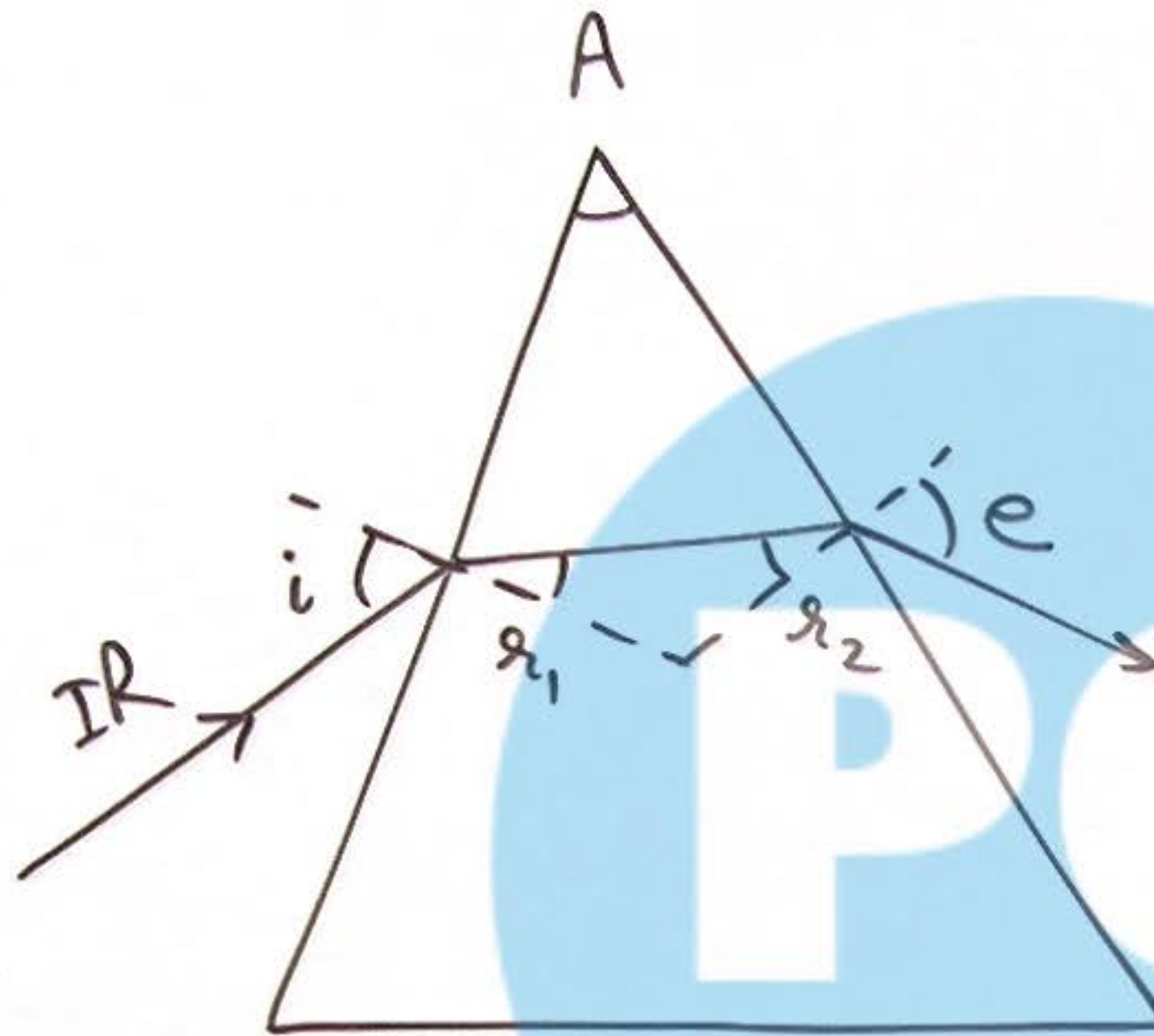
$$\underline{r_2 < \theta_c}$$

$$r_2 < \theta_c$$

$$(A - r_1) < \theta_c$$

max possible value of r_1 is $\underline{\theta_c}$

$\Rightarrow (A - \theta_c) > \theta_c \Rightarrow$ light will NOT pass through prism.
(Min r_2)



If $\underline{\underline{A > 2\theta_c}}$

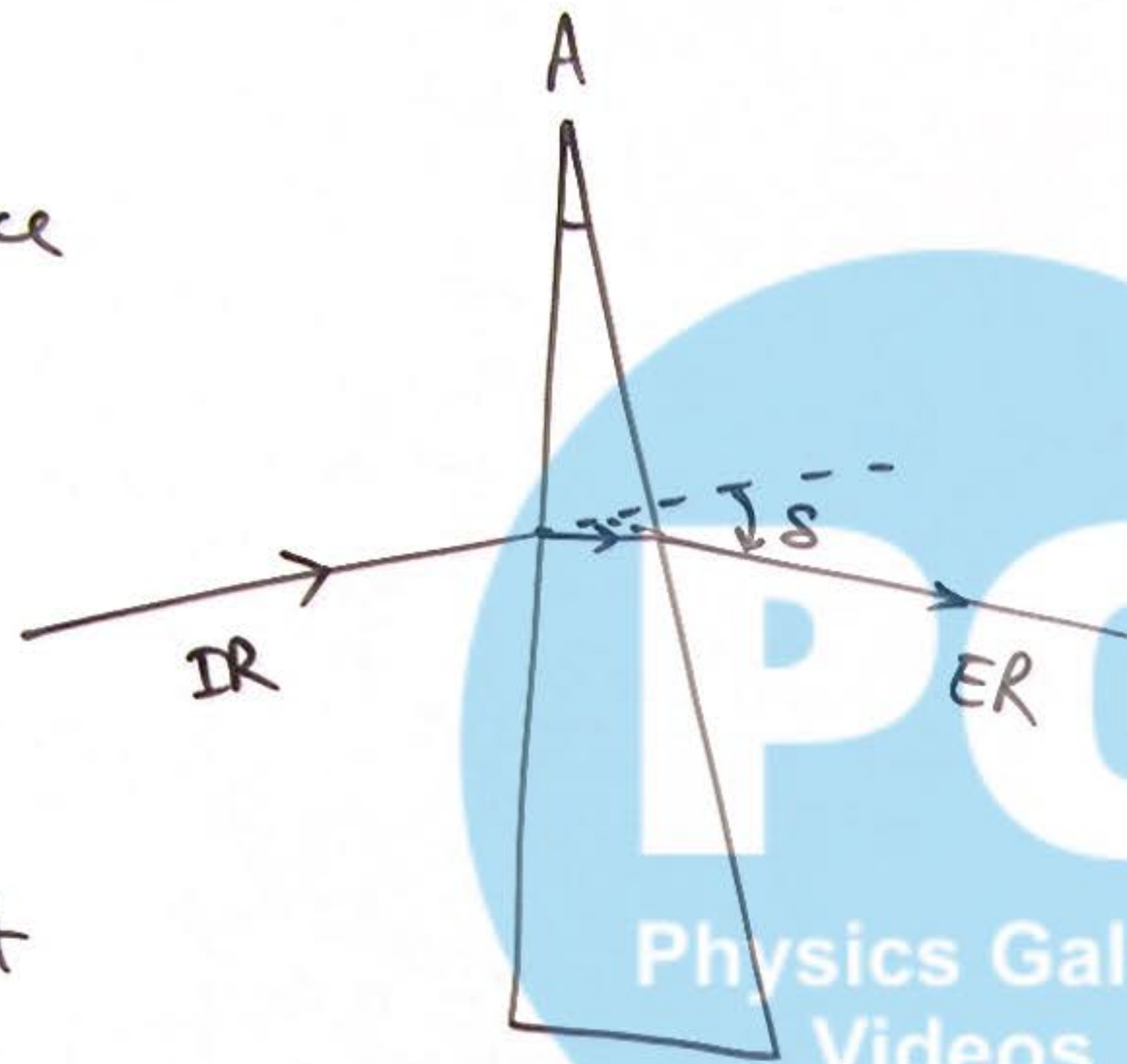
then light will not pass through the prism by total surface Refl.

23) Refraction by a Small Angled Prism:

for almost
normal incidence



Symmetric
passage of
light



$$\sin\left(\frac{A+\delta}{2}\right) = \mu \sin\left(\frac{A}{2}\right)$$

for small angle $\sin \theta \approx \theta$

$$\Rightarrow \frac{A+\delta}{2} = \mu \frac{A}{2}$$

$$\boxed{\delta = A(\mu - 1)}$$

24) Refraction through a Multihedral Prism:

